False Discovery Rate (FDR) in Mass Spectrometry-Based Proteomics

Analysis of Protein Expressions
Project 1
2024-2025

2nd year Master of Statistics and Data Science
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Submission Date: 20 December 2024

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1 Introduction

Proteomic analysis begins by breaking proteins into smaller peptides using enzymatic digestion. These peptides are then separated through liquid chromatography and analyzed using tandem mass spectrometry (LC-MS/MS), which generates peptide fragmentation spectra. Search engines compare these spectra to theoretical ones, resulting in peptide-spectrum matches (PSMs) that help identify the peptides [4]. While this method is effective, it can produce false positives, especially in large datasets. To reduce errors and improve reliability, researchers use the false discovery rate (FDR) as a statistical measure [3].

The most common way to control the false discovery rate (FDR) is by using the target-decoy strategy. This can be done in two ways: the concatenated method, where target and decoy sequences are combined into one search database, or the unconcatenated method, where target and decoy searches are performed separately. Both methods help separate true identifications from random matches, increasing confidence in the results. However, they use different approaches, which can result in different FDR estimates. One of the main goals of this study is to compare these two methods and understand the differences between their FDR results [3]. This study also explored several factors that might affect the accuracy of peptide identification, including peptide length, mass, and charge state. Additionally, this study investigated whether it is sufficient to only include the first-ranked PSM in the spectra information or to also consider the second-ranked PSM.

2 Methodology

2.0.1 Data description

The data comes from a 2015 study by the Association of Biomolecular Resource Facilities (ABRF) which is about the detection of differentially abundant proteins in label-free quantitative LC-MS/MS experiments. It consists of more than 50,000 spectra derived from samples with piked-in standard proteins in a background of *Saccharomyces cerevisiae* or brewer's yeast[1]. A relevant FASTA database for this species was also downloaded from proteomeXchange ftp-site which was used in creating target and decoy database with concatenated and unconcatenated formats.

2.0.2 Softwares Used

The raw file was converted to a mzML format using ProteoWizard 3.0.21229 and the protein identification was performed using SearchGUI 4.3.11 using the Comet search algorithm. Meanwhile, the analyses and visualizations were performed using R programming language.

2.1 SearchGUI and COMET Algorithm

This study used COMET algorithm paired with SearchGUI for identifying peptides and proteins from tandem mass spectrometry (MS/MS) data. This algorithm works by comparing experimental spectra to theoretical spectra generated from protein sequence databases. COMET uses scoring functions, such as the cross-correlation (XCorr) score, to measure how closely the observed and theoretical spectra match. This involves generating theoretical spectra for potential peptides and aligning them with experimental spectra to determine the best peptide-spectrum match (PSM) [2].

2.2 FDR Calculation

False discovery rate (FDR) estimation uses different strategies for the target-decoy concatenated and separate (unconcatenated) methods. In the concatenated method, the target and decoy

sequences are combined into a single database and searched together. The FDR is calculated as:

$$\text{FDR} = \frac{2N_d}{N_t + N_d}$$

where N_d is the number of decoy matches and N_t is the number of target matches. This approach is computationally efficient but can slightly underestimate the FDR because of peptide competition, where decoy matches sometimes score higher than true target matches, reducing the number of correct target identifications.

In the separate (unconcatenated) method, the target and decoy databases are searched independently. This method takes a more conservative approach, and the FDR is calculated as:

$$\text{FDR} = \frac{N_d}{N_t}$$

Here, N_d is the number of decoy matches, and N_t is the number of target matches. This approach avoids peptide competition but may overestimate FDR since some spectra that could match target peptides are instead assigned to decoy peptides [3].

2.3 BLAST

The Basic Local Alignment Search Tool (BLAST) is widely used for identifying regions of similarity between biological sequences, providing insights into sequence relationships and statistical significance. In this study, BLAST was used to directly compare sequences between the top-ranking decoy and target peptides, as well as between the first and second-ranking peptides. This comparison helps evaluate whether focusing only on the top-ranking peptide-spectrum match (PSM) is reliable, or whether lower-ranking matches could be misleading. Additionally, examining the similarity between high-scoring decoy and target peptides aimed to identify potential issues with the randomness of the decoy database. If decoy sequences closely resemble target peptides, it could make the FDR estimation overly conservative, as high-scoring decoy matches may interfere with confident peptide identifications. To ensure consistency, alignments were performed using the BLOSUM62 scoring matrix, which is standard for BLAST, but post-translational modifications (PTMs) were excluded to avoid scoring inconsistencies. This approach ensures a fair evaluation of peptide alignments and highlights the importance of both database design and ranking reliability in FDR filtering and peptide identification.

2.4 Global FDR

In this study, the Global False Discovery Rate (FDR) was evaluated by conducting three separate searches: two for the unconcatenated target and decoy files, and one for the concatenated file, using SearchGUI along with COMET algorithm. The raw data was loaded into SearchGUI, and the search parameters included trypsin as the enzyme, a maximum of two missed cleavages, a fragment mass tolerance of 0.02 Da, precursor mass tolerance of 10 ppm, and peptide charges of 2+, 3+, and 4+. Modifications such as oxidation and phosphorylation were also considered. To calculate the FDR, rank-1 PSMs were used. We examined how FDR varied with score thresholds, charge state, peptide length, and peptide mass.

2.5 Local FDR

Based on the previous results of SearchGUI, the spectrum or scan numbers from the 100 best scoring PSMs in the target database were noted and used to filter the mzML database. This new dataset was used to search not only for the rank 1 PSM but also the top 100 for target and decoy database separately. A total of 100 spectra $\times$ 100 PSM = 10,000 PSMs were generated from each database, target and decoy. The score distributions between the first and second ranking peptides were analyzed using distribution plots and the similarities between the two best matching peptides are further analyzed using BLAST global alignment.

3 Results and Discussion

3.1 Global FDR

3.1.1 Score Distribution

In the separate target-decoy (unconcaatenated) search, the target and decoy peptide sequences are clearly separated in terms of Xcorr values as shown in Figure **??**. The target peptides have higher Xcorr values, and the decoy peptides typically have lower scores. The decoy peptides are often random sequences designed to mimic the targets, so they should have lower Xcorr scores. Meanwhile, in the concatenated database, both target and decoy peptides are mixed in the same dataset. This mixing leads to more overlap between the target and decoy distributions, especially in the lower Xcorr range. As a result, distinguishing targets from decoys based purely on Xcorr becomes more challenging, especially for lower score thresholds.

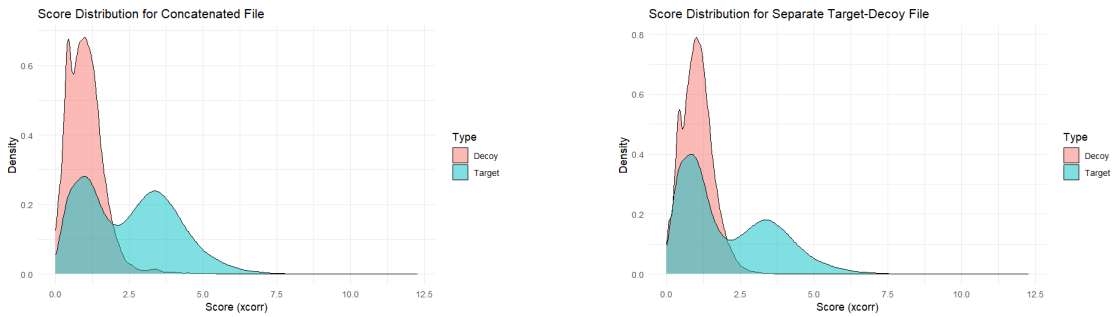


Figure 1: Score distributions for concatenated (left) and unconcatenated (right) database

3.1.2 FDR versus Score

As shown on Figure 2, for both concatenated and unconcatenated database, as we increase the score threshold (increasing Xcorr), the FDR decreases, meaning the proportion of decoy peptides that are mistakenly identified as target peptides decreases.
For the separate target-decoy database, the sharp drop in FDR occurs at a lower threshold, suggesting that it is easier to distinguish between target and decoy peptides because their Xcorr distributions are well separated. When the score threshold is set high, very few decoy peptides pass the threshold, resulting in a low FDR.

In contrast, for the concatenated database, the FDR remains higher for the same score threshold because the distributions of target and decoy peptides are overlapping. This results in some decoy peptides scoring highly enough to be mistaken for target peptides, particularly at lower thresholds of Xcorr.

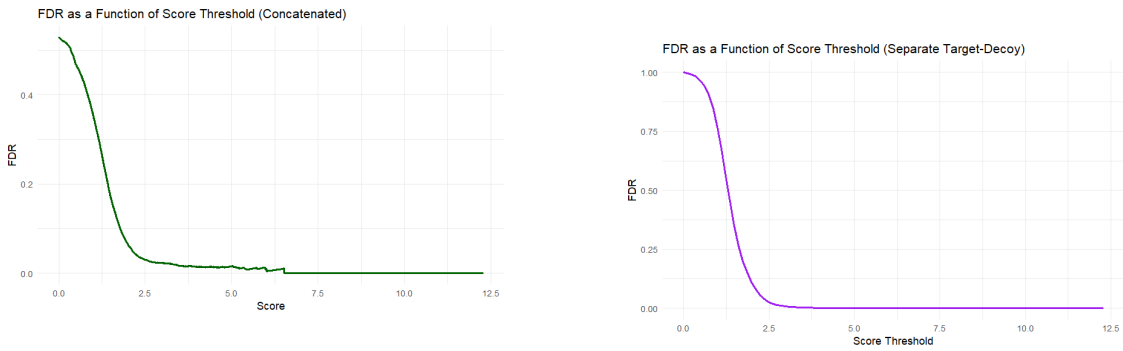


Figure 2: FDR versus score in concatenated (left) and unconcatenated (right) database

3.1.3 Rank-1 Decoy vs Rank-1 Target Score

Figure 3 shows a histogram on the pairwise alignment scores between rank-1 target and rank-1 decoy peptides. It can be seen that majority have low similiarity scores with values peaking at around 10 which suggests that these two type of peptides beign compared are not very similar.

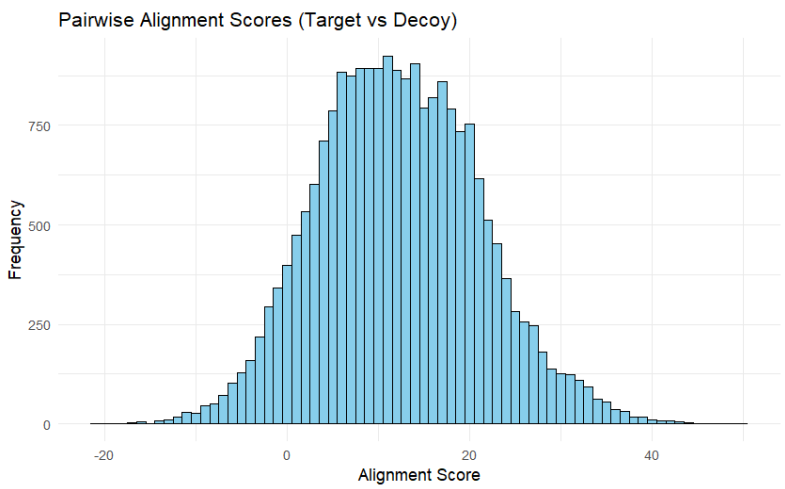


Figure 3: BLAST scores distribution for rank-1 decoy versus rank-1 target PSMs

3.2 Effect of Charge State, Length, and Mass

The cross-correlation (Xcorr) scores grouped by charge state is shown in FIgure **??**. The Xcorr scores are relatively consistent across different charge states, with peptides having charge states 2+, 3+, and 4+ showing some variation. Peptides with higher charge states (3+ and 4+) exhibit a slightly larger variation in their cross-correlation scores compared to those with a 2+ charge state. This suggests that peptides with higher charge states might be fragmented differently, resulting in slightly different Xcorr score distributions. A higher charge state can lead to more intense and clearer fragment ion signals, which may explain the variation observed in these charge states.

Figure 5 is showing the distribution of peptide mass in the dataset, specifically how the mass of peptides correlates with the Xcorr scores. The majority of peptides fall within a mass range of 1000 to 2000 Da, and the cross-correlation scores for these peptides are typically higher compared to very light or very heavy peptides. Lighter peptides may have fewer fragment ions to compare, leading to less reliable identification and lower xcorr scores. Conversely, very large peptides might not fragment as efficiently, leading to similar issues with score reliability. The trend suggests

Figure 4: Distribution of charge states

that peptides within the 1000-2000 Da range provide the most reliable results in terms of cross-correlation and peptide identification.

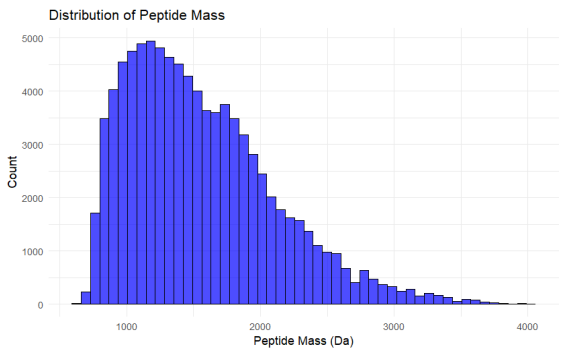
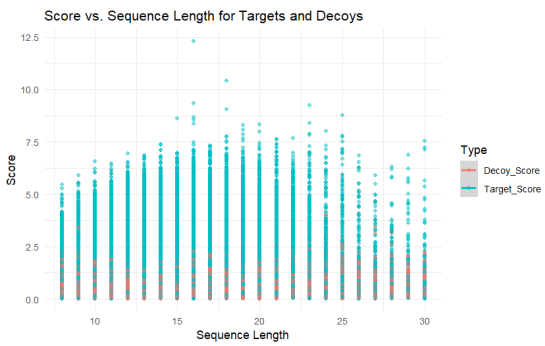
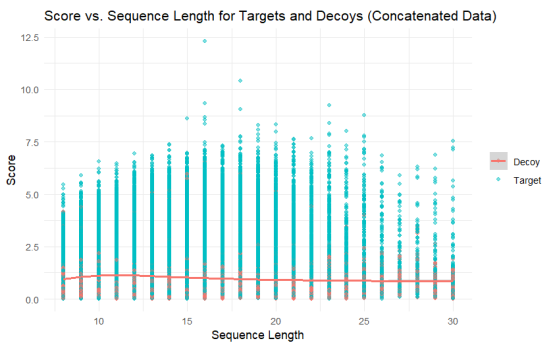


Figure 5: Distribution of peptide mass

Figure 6 shows the relationship between peptide length and cross-correlation score. The xcorr scores remain relatively consistent across peptides of different lengths, with a slight increase in score for medium-length peptides (between 10-20 amino acids). This can be explained by the fact that peptides of medium length tend to have more detectable fragment ions, which can generate more accurate and confident cross-correlation scores. However, longer or shorter peptides may face difficulties in fragmentation or accurately identifying the correct matching ions, which can lead to slightly lower average cross-correlation scores.



(a) Sequence length versus score in unconcatenated database

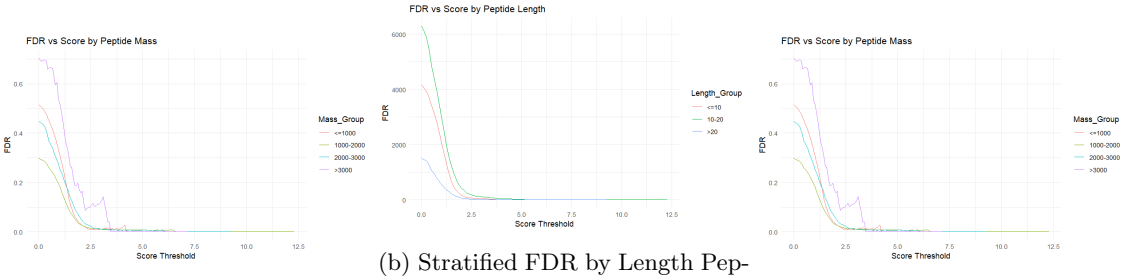


(b) Sequence length versus score in concatenated database

Figure 6: Sequence length versus score

3.3 Statistical assessment of stratified FDR

As shown in Figures 7, varying FDR levels across different peptide lengths, masses, and charge states suggest that treating these factors separately may lead to more accurate and reliable identification thresholds. Therefore, we further analyzed our assumption, by testing it with by Kruskal-Wallis test.



(a) Stratified FDR by charge

(b) Stratified FDR by Length Peptide

(c) Stratified FDR by Mass

Figure 7: Stratified FDR evolution

To evaluate the differences in X-corr values across charge states, we used the Kruskal-Wallis rank sum test to analyze our data and the results are shown on Table 1. The test revealed a statistically significant difference among the charge states, with a p-value of < 0.0001. Based on this result, we proceeded with pairwise Wilcoxon tests to examine differences between individual charge state pairs. The analysis showed significant differences between charge states 3 and 4, as well as 2 and 4. However, no significant difference was observed between charge states 2 and 3.

Table 1: P-values from pairwise comparisons of charge states.

Charge State 1	**Charge State 2**	**p-value**
2	3	0.42
3	4	< 0.0001
4	2	< 0.0001

For each charge state (2+, 3+, and 4+), we calculated the cross-correlation (xcorr) threshold that corresponds to an FDR near 1% or slightly below. The summary is displayed on Table 2. These FDR values reflect the probability that the identified matches are true target peptides as opposed to false decoy peptides. Specifically, for charge state 2+, an FDR of approximately 0.97% is achieved at an xcorr threshold of 2.36, for charge state 3+ at 3.60 xcorr, and for charge state 4+ at 4.02 xcorr, where the FDR drops to about 0.89%. Notably, as the charge state increases, the Xcorr threshold needed to reach a 1% FDR also rises. This trend is consistent with proteomics, where peptides with higher charge states typically require more stringent cross-correlation thresholds to ensure accurate peptide identification.

Table 2: Cross-Correlation thresholds and corresponding FDRs for different charge states

Charge State	**Xcorr Threshold**	**FDR**
2+	2.363889	0.009729473
3+	3.599929	0.009737098
4+	4.016465	0.008928571

The Kruskal-Wallis test for FDR by peptide length groups revealed a significant difference (p-value = 0.00143), indicating that FDR values vary significantly across different peptide lengths. Therefore, we moved further and observed pairwise comparisons revealing that FDR values be-

tween length groups ≤10 and >20 showed significant differences (p-value = 0.00075), while other comparisons did not reach significance.

Comparison	p-value
Length $\leq$ 10 vs Length 10-20	0.05615
Length 10-20 vs Length > 20	0.00075
Length $\leq$ 10 vs Length > 20	0.18939

Table 3: Pairwise Comparisons for FDR by Peptide Length Groups using Wilcoxon Test

The Kruskal-Wallis test for FDR by peptide mass groups showed a highly significant result (p-value = 2.24e-05), indicating that FDR differs significantly across peptides of different mass ranges. Pairwise comparisons for FDR by peptide mass groupe revealing significant differences between mass groups ≤1000 and 1000-2000 (p-value = 0.010), and between 1000-2000 and >3000 (p-value = 4.8e-06).

Comparison	p-value
Mass $\leq$ 1000 vs Mass 1000-2000	0.010
Mass 1000-2000 vs Mass 2000-3000	0.010
Mass 1000-2000 vs Mass > 3000	< 0.0001
Mass 2000-3000 vs Mass > 3000	0.033

Table 4: Pairwise Comparisons for FDR by Peptide Mass Groups using Wilcoxon Test

3.4 Local FDR

Figure 8 shows the joint Xcorr score distribution of all PSMs in the target database (left) and decoy database (right). Both distributions show that the majority scores fall between 0 to 3 but more scores between 0 and 1 can be observed in the decoy database. Moreover, some outlier scores exist in target database with high score values while this cannot be seen in the decoy database. The highest Xcorr observed in the decoy database is only 4.421 while for target is 8.672. These observations are to be expected since the target database contains real protein sequences that are more likely to generate meaningful matches with the observed spectra unlike with decoy database where the protein sequences are not real.

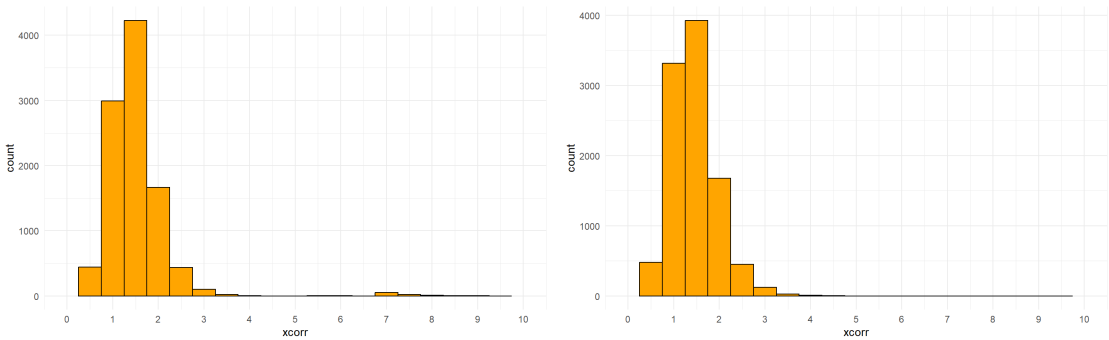


Figure 8: Score distribution for all PSMs in target (left) and decoy (right) database

The score distributions of the 100 spectra for the target and decoy database are shown in Figure 9. For both distributions, differences can be observed between different spectra, some have narrow peak while some are widely distributed. A few spectra distributions from the target database are right-tailed and wider compared to the decoy one. This is driven by the existence of some high scoring peptides in some spectra for the target database which are non-existent in the decoy

database.

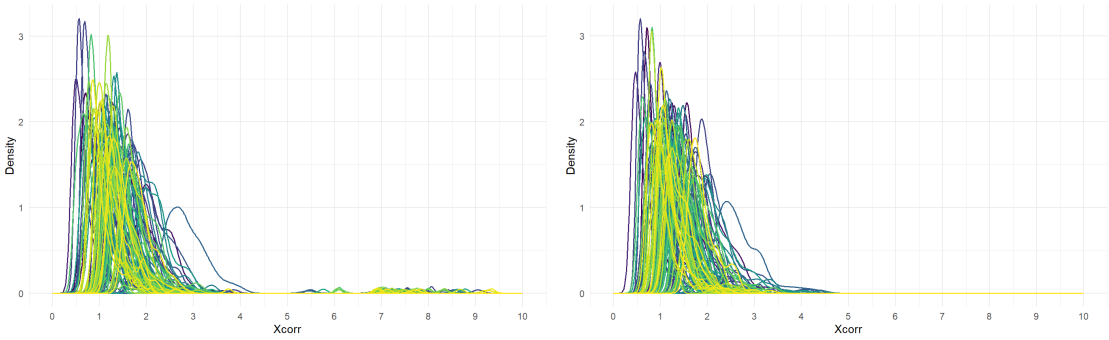


Figure 9: Score distributions of 100 spectra in target (left) and decoy (right) database

To assess the differences between the top scoring peptide and the second best match, delta correlation ΔCn is often used in mass spectrometry-based proteomics. It is calculated as:

$$\Delta Cn = \frac{\text{Xcorr}_{\text{best}} - \text{Xcorr}_{\text{second-best}}}{\text{Xcorr}_{\text{best}}}$$

where the difference in score (Xcorr) between the best match and the second best match is divided by the best match score to standardize the value. A high value, that is close to 1 indicates a large difference between the best and second-best match which suggests that the top scoring peptide is more likely correct because it is clearly better than the next best-match peptide. Figure 10 shows the Delta Cn distribution between rank 1 and rank 2 peptides in the target and decoy database. It can be seen that most of the values are in the 0.5 to 0.8 range which implies many high confidence matches were found in the target database. On the other hand, majority of the matches found in the decoy database are of low-confidence with values peaking in the range from 0 to 0.1. This suggests that it is more difficult to identify the best peptide match in this database. This is to be expected in a decoy database since matches here are inherently less reliable and serve as a control for identifying false positives in the target database.

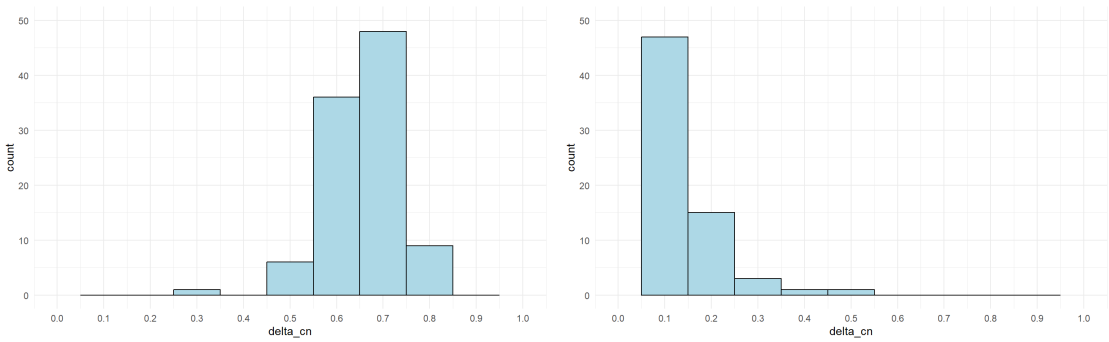


Figure 10: Distribution of Delta Cn between rank 1 and rank 2 peptides in target (left) and decoy (right) database

The similarities between the top 1 and top 2 peptides were further analyzed using BLAST. It is interesting to see the BLAST scores from the spectra with the lowest Delta Cn from the decoy database since low values would imply low-confidence match and the difference between the two peptides being compared would be small or negligible. Table 5 shows the peptide sequences for the best and second best matching peptide for spectra with the five lowest Delta Cn in decoy

database. It can be observed that the sequences between the two peptides are very different for each scan with very low BLAST alignment scores, some being negative. The only similarity that is quite consistent would be the length of the sequences.

Table 5: Top two ranking peptides for spectra with the lowest Delta Cn in decoy database

Scan ID	Rank 1 Peptide	Rank 2 Peptide	Score
16496	SLKVMACNLR	KEATVNDMQR	-5
26414	YGTPLGHLKDGHSMWVTSDNMGK	QYLEKEALSIADKENNLMEVVK	-28
37350	IAEQFLLDPEMQLSMK	IEVNSDPKPDMEPYFK	10
42673	EGNRELACNTALTTGMR	NMITEASFTYDIGFKR	-8

The BLAST score distributions between the top two scoring peptides for the target and decoy database are shown in Figure 11. In both plots, the majority of the scores are low with some having negative values, which is more pronounced in the decoy database. Thus, slightly lower scores can be observed in the decoy database compared to target database. A bit more high scores can also be seen in the target database compared to the decoy database but these high scores are only very few for both databases and can even be considered outliers based on the overall distributions.

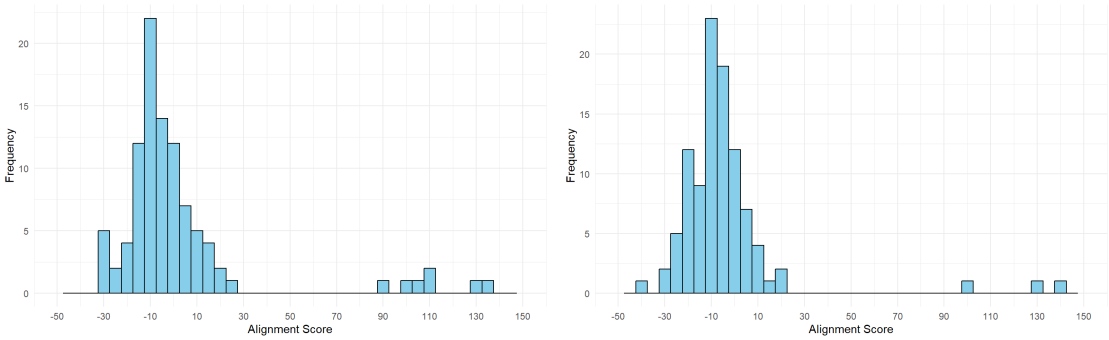


Figure 11: BLAST scores distribution between rank 1 and rank 2 peptides in target (left) and decoy (right) database

4 Conclusion

In the comparison between concatenated and unconcatenated database, more higher score matches coming from the target database are found in the concatenated one while the higher scoring peptides from the decoy are more in the unconcatenated database. This result suggests that the concatenated database seems to better discriminate between true and false matches, whereas the unconcatenated database appears to result in higher confidence scores for false matches. This highlights a potential advantage of the concatenated approach in reducing false positives.

In estimating the local FDR for a peptide match, a search is usually performed against both target and decoy databases, and scores are calculated. It was shown in the results that the score distributions are different for the two database. These two score distributions should have been fitted with a mixture model based on the distributions of both target and decoy database to estimate the local FDR for each peptide match. This, however was not performed in this paper.

In the comparison between the top 1 and top 2 ranking peptides, more high confidence matches were observed in the target database than in the decoy database. The BLAST scores revealed that similarities between rank 1 and rank 2 peptides are rare for the two database. In cases where there are no similarities between the two top ranking peptides and the rank 1 peptide has a poor

score, it is possible to miss peptides that are actually correct but ranked lower, resulting in a loss of sensitivity. The rank 1 peptides could potentially be better matches in cases where the rank 1 match is spurious so ignoring them means missing valuable true positives. Therefore, it is also important to consider the second rank peptides if the goal is to maximize sensitivity. However, we often want to balance sensitivity with specificity and properly evaluating the second-ranked peptides can help minimize false positives and maintain an acceptable FDR.

References

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[2] Jimmy K Eng, Michael R Hoopmann, Tahmina A Jahan, Jarrett D Egertson, William S Noble, and Michael J MacCoss. A deeper look into Comet—Implementation and features. *Journal of The American Society for Mass Spectrometry*, 26(11):1865–1874, November 2015.

[3] Alexey I. Nesvizhskii. A survey of computational methods and error rate estimation procedures for peptide and protein identification in shotgun proteomics. *Journal of Proteomics*, 73(11):2092–2123, 2010. Model organism proteomics.

[4] Fang Xie, Tao Liu, Wei-Jun Qian, Vladislav A Petyuk, and Richard D Smith. Liquid Chromatography-Mass spectrometry-based quantitative proteomics *. *Journal of Biological Chemistry*, 286(29):25443–25449, July 2011.

Appendix

R Codes

```
#Global FDR

ggplot(separate_combined, aes(x = Score, fill = Type)) +
  geom_density(alpha = 0.5) +
  labs(title = "Score Distribution for Separate Target-Decoy File",
       x = "Score (xcorr)", y = "Density",
       fill = "Type") +
  theme_minimal()

ggplot(concat_combined, aes(x = Score, fill = Type)) +
  geom_density(alpha = 0.5) +
  labs(title = "Score Distribution for Concatenated File",
       x = "Score (xcorr)", y = "Density",
       fill = "Type") +
  theme_minimal()

ggplot(separate_data, aes(x = Target_Score)) +
  geom_density(fill = "blue", alpha = 0.5) +
  geom_density(aes(x = Decoy_Score), fill = "red", alpha = 0.5) +
  labs(title = "Score Distribution for Target and Decoy PSMs",
       x = "Score (xcorr)", y = "Density") +
  theme_minimal()

fdr_separate <- data.frame(Threshold = score_thresholds) %>%
  rowwise() %>%
  mutate(
    ND = sum(separate_data$Decoy_Score >= Threshold, na.rm = TRUE),
    NT = sum(separate_data$Target_Score >= Threshold, na.rm = TRUE),
    FDR = ifelse(NT == 0, 0, ND / NT)
  )

ggplot(fdr_separate, aes(x = Threshold, y = FDR)) +
  geom_line(color = "purple", size = 1) +
  labs(title = "FDR as a Function of Score Threshold (Separate Target-Decoy)",
       x = "Score Threshold", y = "FDR") +
  theme_minimal()

fdr_concat <- concat_data %>%
  arrange(desc(Score)) %>%
  mutate(
    ND = cumsum(Label == "Decoy"),
    NT = cumsum(Label == "Target"),
    FDR = 2 * ND / (ND + NT)
  )

ggplot(fdr_concat, aes(x = Score, y = FDR)) +
  geom_line(color = "darkgreen", size = 1) +
  labs(title = "FDR as a Function of Score Threshold (Concatenated)",
       x = "Score", y = "FDR") +
```

```
  theme_minimal()
# Initialize results
alignment_results <- data.frame(Query_ID = character(), Alignment_Score = numeric())

# Loop through each Query_ID where Decoy scores are higher
for (i in 1:nrow(decoy_higher)) {
  query_id <- decoy_higher$Query_ID[i]
  subject_id <- decoy_higher$Subject_ID[i]

# Match Query_ID to Target FASTA and Subject_ID to Decoy FASTA
  target_match <- names(target_fasta) == query_id
  decoy_match <- names(decoy_fasta) == subject_id

# Extract sequences if matches are found
  if (any(target_match) && any(decoy_match)) {
    target_seq <- as.character(target_fasta[target_match])
    decoy_seq <- as.character(decoy_fasta[decoy_match])

# Perform pairwise alignment
    alignment <- pairwiseAlignment(target_seq, decoy_seq,
                                   substitutionMatrix = "BLOSUM62",
                                   gapOpening = 10, gapExtension = 1)

# Append results
    alignment_results <- rbind(alignment_results,
                               data.frame(Query_ID = query_id, Alignment_Score = score(alignment))
  } else {
    cat("Warning: No match found for Query_ID:", query_id, "or Subject_ID:", subject_id, "\n")
  }
}

# Print and summarize alignment results
print(head(alignment_results))
summary(alignment_results$Alignment_Score)

# Plot the alignment score distribution
library(ggplot2)
ggplot(alignment_results, aes(x = Alignment_Score)) +
  geom_histogram(binwidth = 1, fill = "skyblue", color = "black") +
  labs(title = "Pairwise Alignment Scores (Target vs Decoy)",
       x = "Alignment Score", y = "Frequency") +
  theme_minimal()

ggplot(combined_data_vis, aes(x = factor(charge), y = Score, fill = Dataset)) +
  geom_boxplot() +
  scale_fill_manual(values = c("Concatenated" = "lightblue", "Decoy" = "lightgreen",
  "Target" = "lightcoral")) +
  labs(title = "Distribution of Scores by Charge State",
       x = "Charge State", y = "Score") +
  theme_minimal() +
  theme(legend.title = element_blank())
```

```
score_length <- separate_data %>%
  select(Target_Length, Target_Score, Decoy_Score) %>%
  pivot_longer(cols = c(Target_Score, Decoy_Score), names_to = "Type",
  values_to = "Score") %>%
  filter(!is.na(Score))

ggplot(score_length, aes(x = Target_Length, y = Score, color = Type)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "loess") +
  labs(title = "Score vs. Sequence Length for Targets and Decoys",
       x = "Sequence Length", y = "Score") +
  theme_minimal()

ggplot(concat_data, aes(x = Length, y = Score, color = Label)) +
  geom_point(alpha = 0.5) +  # Points with transparency
  geom_smooth(method = "loess") +  # LOESS smoothing line
  labs(title = "Score vs. Sequence Length for Targets and Decoys (Concatenated Data)",
       x = "Sequence Length", y = "Score") +
  theme_minimal() +  # Minimal theme for a cleaner look
  theme(legend.title = element_blank()) # Remove legend title

concat_data$Length_Group <- cut(concat_data$Length, breaks =
c(0, 10, 20, Inf), labels = c("<=10", "10-20", ">20"))
concat_data$Mass_Group <- cut(concat_data$mass, breaks = c(0, 1000, 2000, 3000, Inf),
labels = c("<=1000", "1000-2000", "2000-3000", ">3000"))

# Step 2: Calculate FDR for each peptide group at various thresholds
calculate_fdr_thresholds <- function(data, score_column, label_column) {
# Create a sequence of thresholds for score values
score_thresholds <- seq(min(data[[score_column]], na.rm = TRUE), max(data[[score_column]],
na.rm = TRUE), length.out = 100)

# Initialize a data frame to store FDR results
fdr_results <- data.frame(Score_Threshold = score_thresholds,
FDR = numeric(length(score_thresholds)))
# Loop over each threshold to calculate FDR
for (i in seq_along(score_thresholds)) {
    threshold <- score_thresholds[i]

# Count targets and decoys above the threshold
target_count <- sum(data[[label_column]] == "Target" & data[[score_column]] >= threshold)
decoy_count <- sum(data[[label_column]] == "Decoy" & data[[score_column]] >= threshold)

# Calculate FDR (decoy_count/target_count)
    if (target_count > 0) {
      fdr_results$FDR[i] <- 2*decoy_count / target_count+decoy_count
    } else {
      fdr_results$FDR[i] <- NA
    }
  }

  return(fdr_results)
```

```
}

# Step 3: Calculate FDR for different length and mass groups
fdr_by_length <- concat_data %>%
  group_by(Length_Group) %>%
  do(calculate_fdr_thresholds(., "Score", "Label"))

fdr_by_mass <- concat_data %>%
  group_by(Mass_Group) %>%
  do(calculate_fdr_thresholds(., "Score", "Label"))

# Step 4: Plotting FDR vs Score for Length and Mass Groups
ggplot(fdr_by_length, aes(x = Score_Threshold, y = FDR, color = Length_Group)) +
  geom_line() +
  labs(title = "FDR vs Score by Peptide Length", x = "Score Threshold", y = "FDR") +
  theme_minimal()

ggplot(fdr_by_mass, aes(x = Score_Threshold, y = FDR, color = Mass_Group)) +
  geom_line() +
  labs(title = "FDR vs Score by Peptide Mass", x = "Score Threshold", y = "FDR") +
  theme_minimal()

kruskal_test <- kruskal.test(xcorr ~ factor(charge), data = concat_clean)

print(kruskal_test)

pairwise_comparison <- pairwise.wilcox.test(concat_clean$xcorr, concat_clean$charge)

# Display the result of the pairwise comparison
print(pairwise_comparison)

# Kruskal-Wallis test for FDR by peptide length groups
kruskal_length <- kruskal.test(FDR ~ Length_Group, data = fdr_by_length)
print(kruskal_length)

# Kruskal-Wallis test for FDR by peptide mass groups
kruskal_mass <- kruskal.test(FDR ~ Mass_Group, data = fdr_by_mass)
print(kruskal_mass)

# If Kruskal-Wallis test is significant, perform pairwise comparisons
# For length groups
pairwise_length <- pairwise.wilcox.test(fdr_by_length$FDR, fdr_by_length$Length_Group, p.adjust.me
print(pairwise_length)

# For mass groups
pairwise_mass <- pairwise.wilcox.test(fdr_by_mass$FDR, fdr_by_mass$Mass_Group,
p.adjust.method = "BH")
print(pairwise_mass)

# Local FDR
ggplot(target_top100, aes(x = xcorr)) +
  geom_histogram(binwidth = 0.5, fill = "orange", color = "black") +
```

```
  scale_x_continuous(limits = c(0, 10), breaks = seq(0, 10, by = 1)) +
  theme_minimal()

ggplot(decoy_top100, aes(x = xcorr)) +
  geom_histogram(binwidth = 0.5, fill = "orange", color = "black") +
  scale_x_continuous(limits = c(0, 10), breaks = seq(0, 10, by = 1)) +
  theme_minimal()

ggplot(target_top100, aes(x = xcorr, color = factor(scan))) +
  geom_density(alpha = 0.6, size = 0.7) +
  scale_color_viridis_d() +
  labs(x = "Xcorr", y = "Density", color = "Scan") +
  scale_x_continuous(limits = c(0, 10), breaks = seq(0, 10, by = 1)) +
  theme_minimal() +
  theme(legend.position = "none")  #

ggplot(decoy_top100, aes(x = xcorr, color = factor(scan))) +
  geom_density(alpha = 0.6, size = 0.7) +
  scale_color_viridis_d() +
  labs(x = "Xcorr", y = "Density", color = "Scan") +
  scale_x_continuous(limits = c(0, 10), breaks = seq(0, 10, by = 1)) +
  theme_minimal() +
  theme(legend.position = "none")

alignments_decoy <- list()
# Loop through each sequence pair
for (i in 1:nrow(decoy_100_rank1)) {
    seq1 <- AAString(decoy_100_rank1$plain_peptide[i])  # Sequence from dataset 1
    seq2 <- AAString(decoy_100_rank2$plain_peptide[i])  # Sequence from dataset 2

 Perform pairwise alignment
    alignments_decoy[[i]] <- pairwiseAlignment(seq1, seq2,
    substitutionMatrix = "BLOSUM62",
    gapOpening = 0, gapExtension = 8)

# Plot the histogram of alignment scores
ggplot(summary_results_decoy, aes(x = Alignment_Score_decoy)) +
    geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
    labs(
        x = "Alignment Score",
        y = "Frequency"
    ) +
  scale_x_continuous(limits = c(-50, 150), breaks = seq(-50, 150, by = 20)) +
    theme_minimal()
```